

LYRA-CPP — THETIS DIRECT-PORT PLAN (TX)

Authoritative doc for the current TX work direction.

[!] The older EXECUTION_PLAN.md (Step 14 wire-layer rebuild, Phase 3 etc.) describes a DIFFERENT approach that is now superseded for the TX work. Do NOT mix the two plans — that doc tracks an earlier wire-rebuild pass which was partial and Lyra-native-leaning. THIS doc is the active plan going forward. Old doc remains in tree for historical reference only (some shipped pieces from it are still in production — noted per-component in the registry below).

Branch: tx-rebuild (the direct-port work lives here; origin/main carries the earlier Step 14 / TX-1 line and is deliberately untouched by the direct-port arc).

Reference tree (the ONLY trusted TX reference):

- D:\sdrprojects\OpenHPSDR-Thetis-2.10.3.13\Project Files\Source\
 - wdsp*.c — WDSP DSP engine (GPL v3+, port-with-attribution OK)
 - ChannelMaster*.c — wire layer (GPL v3+, port-with-attribution OK)
 - Console*.cs — C# UI/control glue (study ONLY; write Lyra-native equivalents)

- Y:\Claude local_hl2src\ — HL2+ ak4951v4 gateway RTL

(control.v / radio.v / usopenhpsdr1.v etc. — the ground truth when host-side reference and gateway disagree)

Provenance & attribution rule (operator-locked 2026-06-09, supersedes the 2026-05-31 "no-attribution-in-commits" tightening):

Lyra-cpp is a derivative work of openHPSDR Thetis (ChannelMaster + WDSP modules), GPL v3+. The GitHub project description openly states this: **"TX baseline ported from openHPSDR Thetis (ChannelMaster) with Lyra-native DSP additions."** Open attribution is both legally required (GPL) and operator-intended (public project posture).

1. **License attribution at file head — REQUIRED.** Any file ported from WDSP or ChannelMaster carries a header block crediting upstream (NR0V for WDSP modules; openHPSDR Thetis for ChannelMaster modules) + the GPL v3+ notice. Existing AAMix.{h, cpp}, CMaster.cpp, RadioNet.cpp etc. already follow this; keep it.
2. **Code comments citing source — ENCOURAGED.** Citations may name the reference openly (Reference: Thetis ChannelMaster cmaster.c:297-313, // Per openHPSDR networkproto1.c:1078, // HL2+ ak4951v4 gateway control.v:209-220). The file:line is the primary technical content; the reference-app name is context that helps future readers find the source tree.
3. **Commit messages — REFERENCE-APP NAMES PERMITTED.** A commit like **"port aamix.c from Thetis ChannelMaster with Lyra-native idiom translations"** is fine — it matches the public GitHub posture and makes the history grep-able. The earlier 4cef88d tightening (2026-05-31) that put commit messages on the no-attribution list is **rescinded**.
4. ****Public docs (README, docs/*.md design docs, memory files, this plan) — OPEN ATTRIBUTION.**** Same posture as the GitHub description.
5. **User-facing UI strings — KEEP OPERATOR-FOCUSED.** UI labels operators see in normal operation (panel titles, button text, error toasts) describe Lyra behavior, not provenance. **"AAMix output (HL2 jack)"** not **"Thetis-porting AAMix output"**. About dialog / credits screen / install docs MAY (and probably should) name openHPSDR Thetis + WDSP/NR0V openly — credit where due. This is the only category that stays "Lyra-native voice in operator-facing labels," because operators interact with Lyra, not its source provenance.
6. **The "Lyra-Native style governs surrounding architecture only" rule (Stage B-locked) STANDS unchanged.** Reference DSP API call patterns are byte-for-byte faithful; Qt/Vulkan/QML/process model is Lyra-native.
7. **Reference precedence STANDS unchanged.** Verify TX ONLY against Thetis 2.10.3.13 (D:\sdrprojects\OpenHPSDR-Thetis-2.10.3.13\...) + the HL2+ ak4951v4 gateway RTL (Y:\Claude local_hl2src\). **Old**

Python Lyra is NOT a trusted TX reference (its TX never worked).

Consequence of the 2026-06-09 amendment: the three historical commits (efdde02, 604d87b, ed9f0cd) that "predated the no-attribution rule" and contained "Thetis" in their messages are **no longer in violation** — they're now consistent with the rule. Task #73 ("Cleanup pre-existing 'Thetis' mentions in shipped code") is **OBSOLETE**d by this amendment.

LOCKED METHODOLOGY (Stage B-validated, do NOT relax)

These rules were earned through the Stage B aamix.c port arc (2026-06-08) — see SESSION_LOG.md for the full trail.

1. ****Smallest revertable empirical step -> operator HL2 bench ->**

next step.** Each commit is one focused change with a falsifiable bench gate. Multi-step "design then implement" is for stages that genuinely need it (operator approves); default is iterate-and-bench.

2. **"Do as Thetis does, Lyra-Native style"** (locked)

2026-05-30 after the fexchange0 crash arc, re-validated tonight after the B.6.b shortcut attempt cost a revert cycle). Three sub-rules:

- Every WDSP API call matches the reference byte-for-byte

(parameter values, order, the SET of setters fired, the lifecycle stage at which they fire). If tempted to pick a value the reference doesn't use, STOP and find what the reference uses. "I have a reason to pick differently" means I don't. Reference has been right every time.

- "Lyra-Native style" governs surrounding architecture ONLY

(Qt + Vulkan + QML + process/thread model + facade APIs + IPC + UI shell). The WDSP DSP-engine call surface is reference-bound.

- Provenance only in code COMMENTS + design docs + memory.

Never commit messages, never UI strings.

3. **Counters first when the symptom is "nothing happens".**

The Stage B 2-stage instrumentation (chain counters in xMixAudio/mix_main + dispatch counters in dispatchAudioFrame) localized the silent-audio defect to a single line of code on the 2nd bench. Pure speculation would have cost a multi-round revert/speculate cycle. Atomic counter + rate-limited qInfo = trivial overhead, very high diagnostic ROI when "I press play, hear nothing, log looks normal."

4. **Operator-empirical rule overrides agent inference.**

When operator hardware data contradicts what an audit/red-team/Plan-agent concluded, the data wins. Do not defend the audit — re-open, bisect, fix. This rule has fired repeatedly through the broader project arc; the Stage B B.6.b-fix1 fix was an instance (chain counters healthy suggested AAMix was correct; the temptation was to revert the whole port; the empirical pattern instead pointed at the wire-up surface where the bug actually was).

5. ****Read the reference FIRST, in shipped-code form, before**

designing.** Every time we designed first and read second, we shipped a regression. Every time we read first, we got the right answer in one pass. The no-attribution rule applies to shipped code; it does NOT mean "don't read the reference." Read it, dossier the file:line in the plan doc, then write Lyra-cpp.

6. **Session-start discipline:** git fetch origin is the

FIRST action every session, before reading any local state. If divergence exists, surface it to the operator BEFORE writing code. No exceptions.

7. **Push after every component ships, not at EOD.** Each

clean operator HL2 bench -> backup bundle + push immediately. Smaller in-flight delta = smaller surprise blast radius.

8. **Backup bundle naming:** _backups/lyra-cpp-YYYY-MM-DD-.bundle.

git bundle create ... --all (covers all refs).

STATUS REGISTRY — what's done, in-progress, pending

Format: per-file or per-component row. Status tags (ASCII so they render identically in the PDF/DOCX siblings):

- [DONE] DONE + operator-bench-validated
- [WIP] IN-PROGRESS (committed but not yet bench-validated, or partially shipped)
- [PEND] PENDING (planned, not started)
- [N/A] N/A (Lyra-native equivalent already shipped; no direct port planned — see notes column)

Wire layer (ChannelMaster — port directly w/ attribution)

Source	Target (lyra-cpp)	Status	Notes
WDSP exports linkage (OpenChannel/fexchange0/analyzers/PS family)	src/wire/wdspcalls.{h,cpp} (P0.a)	[DONE]	P0.a e0c0f4a 2026-06-09 . The single operator-approved linkage seam: fn-ptr table carrying the WDSP exports' exact names, resolved once from the loaded wdsp.dll. PureSignal entry family complete (verified vs dumpbin). RESAMPLE-typed + flush_resample added at P0.c.
ChannelMaster\cmcomm.h (family common header)	src/wire/cmcomm.{h,cpp} (P0.b -> P0.d)	[DONE]	complex typedef + PORT + verbatim malloc0 (wdsp/utilities.c:37-43) + PI + the reference include surface. P0.d added the opaque verbatim twin typedefs for deferred-subsystem types (ANB/NOB/VOX/TXGAIN/ANALYZERS + DELAY at P2.a); EER completed at P2.a 2026-06-12 — full verbatim struct (wdsp/eer.h:30-49) + the umbrella-mapping note (.cpp files include explicit family headers — an include-list umbrella would be circular under #pragma once).
ChannelMaster\ilv.c (TX I/Q interleaver)	src/wire/ILV.{h,cpp} (P0.b)	[DONE]	P0.b 1f49d98 2026-06-11 . VERBATIM direct port (ilv, *ILV twin typedef, raw Outbound fn ptr, pcm->xmtr[] .pilv setters, malloc0/_aligned_free). Mechanical diff vs ilv.{h,c} = whitespace-only. The earlier src/wdsp/ILV retrofit + its pilv[] bank are retired. scratch/test_ilv ALL PASS.
ChannelMaster\cmbuffs.{h,c} (CMB ring + cm_main pump)	src/wire/CmBuffs.{h,cpp} (P0.d)	[DONE]	P0.d afc7950 2026-06-12 . VERBATIM rewrite (cmb, *CMB, #define CMB_MULT (3), malloc0/_aligned_free; the Phase-C retrofit's calloc/intptr_t/guard deviations removed; pcm->in[] alloc moved back to create_cmastar per reference). Mechanical diff = whitespace-only. [DONE] P0.d operator HL2 RX-regression bench PASSED 2026-06-12 .
ChannelMaster\cmsetup.{h,c} (radio structure + id helpers)	src/wire/cmsetup.{h,cpp} (P0.d, NEW)	[DONE]	P0.d afc7950. VERBATIM: cmMAX* sizing macros (16/4/4/2/32), rxid/txid/sp0id/stype/chid/inid/mixinid/getbuffsize, SetRadioStructure + set_cmdefault_rates. main.cpp config: SetRadioStructure(2,1,1,1,0,...) -> chid(0,0)=0 RX1 / chid(1,0)=1 TX matches the live channel layout. [DONE] Bench PASSED 2026-06-12.

Source	Target (lyra-cpp)	Status	Notes
ChannelMaster\cmaster.{h,c} (FULL: struct + create/destroy + xcmaster + setters)	src/wire/CMaster.{h,cpp} (P0.d)	[DONE]	P0.d afc7950. VERBATIM rewrite: full _cmaster struct (cmaster.h:39-99 incl. the PS surface — out[3]/panalalloc/pgain/peer), cmaster, *CMaster, raw TCI callback fn ptrs, enum AudioCODEC, cm = {0}. Verbatim no-arg create_xmtr/destroy_xmtr (out[0..2] malloc0 + OpenChannel(ch 1) + XCreateAnalyzer(TX disp 1) + create_ilv through the wdspcalls seam — the TxChannel RAIL carve-out is DELETED , src/wdsp/TxChannel.{h,cpp} retired). create_cmater/destroy_cmater verbatim per-stream loops (update[] CS + cmbuffs + in[] for all cmSTREAM streams). xcmaster verbatim (update[] CS restored, real stype/txid/chid, TCI memset restored; fexchange0 + monitor xMixAudio + xilv live). All Sendp*/Set* setters ported. Deferred subsystem lines (dexp/anti-vox/txgain/sidetone/pipe/cmasio/analyzers.c/create_rcvr-body) carried IN PLACE as reference text with DEFERRED tags; eer RESTORED VERBATIM at P2.a 2026-06-12 (create_eer run=0 / destroy_eer / xeer / pSetEER* via the wdspcalls seam — the object exists so the sendOutbound/sendProtocol1Samples peer->run derefs are valid, per cmaster.c:212-224). Sole code accommodation: (char*)" at XCreateAnalyzer. [DONE] Operator HL2 RX-regression bench PASSED 2026-06-12 (RX working on the new per-stream pump/ring layout).
ChannelMaster\obbuffs.c (TX-out ring/seam)	src/wire/ObBuffs.{h,cpp} (P1, NEW)	[WIP]	P1 SHIPPED 2026-06-12 — obbuffs.{h,c} verbatim (obb/*OBB twin typedef, file-scope obp four-alias bank, the reference's own calloc/free 2014-era idioms kept). Mechanical diff: .cpp IDENTICAL; .h sole delta = the include-guard define replaced by #pragma once (documented packaging). Sole deferred line: sendOutbound(id, a->out) in ob_main (network.c:1237 = P2 scope), carried as reference text. TU is DORMANT — no Lyra caller of create_obbuffs until P3/P4; wire bytes unchanged.
ChannelMaster\networkproto1.c (HL2 wire writer)	src/hl2_stream.cpp (LIVE) + src/wire/* (Step 14 pieces)	[WIP]	This is the file the Step 14 plan was attacking. Currently a MIX of Step 14 Lyra-native rewrites + the original hl2_stream.cpp body. P2 = sendOutbound/sendProtocol1Samples fidelity audit of the dormant wire layer; P4 = Wire-LIVE switchover (one commit, full HL2 bench gate).
ChannelMaster\network.c (UDP + keepalive)	src/wire/Ep2SendThread.{h,cpp} + src/wire/Ep6RecvThread.{h,cpp}	[WIP]	Step 14 Lyra-native rewrites shipped (\$5/\$6); could be re-reported direct from reference, OR left as-is if reference-parity audit passes. Operator call (folds into P2/P4).

Source	Target (lyra-cpp)	Status	Notes
ChannelMaster\netInterface.c: :create_rnet	src/wire/RadioNet.cpp::c reate_rnet (LIVE)	[DO NE]	Direct port shipped earlier. The Stage-A SendpOutboundRx no-op stub at line 272 was DELETED 2026-06-08 (B.6.b-fix1, commit 8b8e0da) because it clobbered the AAMix wire-up; future codec-id switching must NOT re-introduce a default registration here. P3 = the outbound registrations (SendpOutboundTx(OutBound) etc.) — MUST register AFTER create_xmtr (the verbatim setters have NO null guards, reference ordering).

WDSP RX chain (port directly w/ attribution)

Source	Target (lyra-cpp)	Status	Notes
ChannelMaster\aamix.{c,h}	src/wire/AAMix.{h,cpp} (P0.c)	[DONE]	P0.c e0f2584 2026-06-11 — VERBATIM re-port (supersedes the 2026-06-08 Stage-B C++23-idiom translation, which the operator rejected as a deviation class). aamix, *AAMIX twin typedef + anonymous slew struct, raw Outbound fn ptr, paamix[] bank private to the .cpp, Win32 primitives direct. Mechanical diff vs aamix.{h,c} = whitespace-only. src/wire/resample.h = the public RESAMPLE ABI verbatim. WdspEngine RX wire-up: capturing lambda -> static member fn + TU-scope context ptr (the reference free-fn+global shape). [DONE] Operator HL2 RX bench PASSED 2026-06-11.
wdsp\firmin.c (filter cores)	(consumed via WDSP cffi)	[N/A]	Used internally by WDSP RXA chain via cdef + dlsym. No source-level port needed (the DLL ships with Lyra-cpp via _native/ per the Python-project pattern; cdef + dlsym in wdsp_native.cpp). WISDOM cache fix shipped on Python side; lyra-cpp's RX chain inherits the same path.
wdsp\wisdom.c	(consumed via WDSP cffi)	[N/A]	Same as firmin.c.

WDSP TX chain (per CLAUDE.md §4.1 port table)

Source	Target (lyra-cpp)	Status	Notes
wdsp\TXA.c (channel scaffold)	(consumed via wire/wdspcalls seam — P0.a/P0.d)	[DONE]	TxChannel class DELETED at P0.d afc7950 — the verbatim create_xmtr in wire/CMaster.cpp opens the WDSP TXA channel itself (OpenChannel/fexchange0/CloseChannel through the wdspcalls seam, cmaster.c:177-190 byte-exact), which removed the runtime-DLL RAIL justification the class existed for. wdsp_native.{h,cpp} still carries the SetTXA*/GetTXAMeter cdefs for the operator-control surface.
wdsp\bandpass.c (bp0/bp1)	(consumed via WDSP cffi — SetTXABandpassFreqs)	[N/A]	The §15.23 trap (SetTXABandpassRun overrides bp1 to a stale state) is documented in CLAUDE.md + project_lyra_cpp_tx.md. cffi-only access is the correct posture.
wdsp\compress.c (TX compressor)	src/wdsp/Compress.{h,cpp} OR cffi binding	[PEND]	Operator choice: direct port (~150 LOC) for v0.2.1 EQ + dynamics ship, OR cffi-only binding if WDSP DLL exports are sufficient. Direct port recommended for parity with Lyra's existing UI control surface.
wdsp\cfcomp.c (5-band CFC)	src/wdsp/CFComp.{h,cpp} OR cffi binding	[PEND]	Same operator-choice. ~600 LOC; one of the heavier ports. Replaced in Python Lyra plan by a Lyra-native Combinator per §15.19; cpp may take the same path (custom Combinator instead of WDSP CFC) — operator design call.
wdsp\osctrl.c (CESSB)	cffi binding (SetTXAosctrlRun)	[PEND]	Likely cffi-only; ~200 LOC port not justified unless operator UI need surfaces.
wdsp\wcpagc.c mode 5 (leveler)	cffi binding (SetTXALeveler*)	[WIP]	cdef already present in wdsp_native.cpp. UI surface exists in TX-1 components on main. Confirm reference-parity at the call-site setter sequence.
wdsp\wcpagc.c mode 5 (ALC)	cffi binding (SetTXAALCMaxGain)	[WIP]	cdef present; SetTXAALCThresh does NOT exist (§15.23-class trap — only MaxGain governs ALC ceiling).
wdsp\eqp.c (parametric EQ)	src/wdsp/EQ.{h,cpp} OR cffi binding	[PEND]	Per CLAUDE.md §4.1 v0.2.1; ~300 LOC. Lyra-native parametric EQ likely (§15.19 lists 3-or-5-band parametric replacing the WDSP 10-band graphic).
wdsp\gen.c (gen0/gen1 generators)	cffi binding (SetTXAPreGen*/SetTXAPostGen*)	[WIP]	gen1 (postgen) used today for TUN tone. gen0 (pregen) cdefs deferred per §15.23 (bench-tooling nicety, not on the signal path).

TX PureSignal (per CLAUDE.md §4.1 v0.3)

Source	Target (lyra-cpp)	Status	Notes
wdsp\iqc.{c,h}	src/wdsp/Iqc.{h,cpp} (~315 LOC)	[PEND]	v0.3. cffi math + Lyra-cpp wrapper for the 5-state PS lifecycle (RUN, BEGIN, SWAP, END, DONE).
wdsp\calcc.{c,h}	src/wdsp/Calcc.{h,cpp} (~1164 LOC)	[PEND]	v0.3. cffi math + Lyra-cpp wrapper for the 8-state PS FSM + 3-state auto-attenuator FSM + PSDialog UI.
wdsp\lmath.c::xbuilder	src/wdsp/PsXBuilder.{h,cpp} (~200 LOC)	[PEND]	v0.3. Cubic-spline coefficient builder, used by Lyra-cpp-side calcc orchestration.
wdsp\delay.c	src/wdsp/DelayLine.{h,cpp} (~80 LOC)	[PEND]	v0.3. TX/feedback time-alignment.

Console (C# — study only, Lyra-native equivalents)

Source	Target (lyra-cpp)	Status	Notes
Console\console.cs::chkMOX_CheckedChanged2	src/ptt.{h,cpp} + src/wdsp_engine.cpp keydown/keyup	[WIP]	Lyra-native FSM shipped on main. Verified against the reference's keydown/keyup ordering invariants (RX-DSP stop on keydown -> wire MOX -> TX-DSP start; keyup TX-DSP off -> mox_delay -> clear MOX -> ptt_out_delay -> RX-DSP restart). No direct port — C# is study-only by license.
Console\console.cs (TR delays)	src/tx/TrSequencing.{h,cpp} (LIVE)	[DONE]	Lyra-native shipped with operator's Thetis-DB-verified defaults (mox=15/rf=50/space_mox=13/ptt_out=5/key_up=10 ms).
Console\console.cs::m_bATTonTX	src/tx/AttOnTxPolicy.{h,cpp} (LIVE)	[WIP]	Lyra-native shipped on main. Reference posture: keydown save per-band step-att + force-31 (when PS-A off OR CW); keyup restore. Implementation verified against console.cs:30293-30327 + :30391-30410.
Console\PSForm.cs	src/ui/PsDialog.{h,cpp} (NEW, v0.3)	[PEND]	v0.3 PureSignal UI. C# study only; write Qt/QML native.
Console\HPSDR\IoBoardHL2.cs	src/hl2_stream.cpp (LIVE)	[WIP]	HL2 I/O quirks already implemented Lyra-native via the protocol-byte verification + gateway-RTL ground-truth pass (CLAUDE.md §15.26).

STAGE INDEX — ordered work plan

P0 verbatim-rewrite arc (2026-06-09 -> current; the ACTIVE plan)

After the operator rejected the 2026-06-09 "rule-#8" retrofit deviations ("I said PORT!"), every TX-ported file is being rewritten VERBATIM (byte-faithful; only documented packaging differences — lyra::wire namespace, PORT-expands-empty, the wdspcalls seam for the statically-linked WDSP exports). Each step ships with a mechanical diff vs the reference + an operator HL2 bench gate.

Step	Status	What	Commit / gate
P0.a	[DONE] DONE	wire/wdspcalls.{h,cpp} — the single approved WDSP linkage seam (exact export names, PS entry family complete)	e0c0f4a 2026-06-09

Step	Status	What	Commit / gate
P0.b	[DONE] DONE	ilv.{h,c} verbatim -> wire/ILV.{h,cpp} + new wire/cmcomm.{h,cpp}	1f49d98 2026-06-11; test_ilv ALL PASS
P0.c	[DONE] DONE	aamix.{h,c} verbatim -> wire/AAMix.{h,cpp} + wire/resample.h RESAMPLE ABI	e0f2584; [DONE] operator HL2 RX bench PASSED 2026-06-11
P0.d	[DONE] DONE	cmbuffs/cmaster/cmsetup verbatim -> wire/CmBuffs/wire/CMaster/wire/cmsetup; full _cmaster struct (PS surface); TxChannel carve-out DELETED	afc7950 2026-06-12; [DONE] operator HL2 RX-regression bench PASSED 2026-06-12
P1	[WIP] SHIPPED	obbuffs.c verbatim -> wire/ObBuffs.{h,cpp} (the TX-out seam; separate TU from cmbuffs; sendOutbound call DEFERRED to P2)	2026-06-12; dormant/wire-inert — optional operator RX smoke; clean build 0 warnings; mechanical diff IDENTICAL
P2	[DONE]	sendOutbound / sendProtocol1Samples fidelity audit of the dormant wire layer — COMPLETE 2026-06-12 (P2.a eer completion + P2.b prn outbound surface verbatim + P2.c wire/Network.cpp sendOutbound verbatim, mechanical diff IDENTICAL; the ObBuffs ob_main hand-off restored). P2.a SHIPPED 2026-06-12 — eer completion (verbatim struct + seam entries + the four CMaster restore sites); kills the null-peer landmine both reference functions deref. NEXT: P2.b prn outbound surface verbatim (outLRbufp/outlQbufp/OutBufp -> reference pointer fields + the HANDLE semaphore quartet; callers bend) -> P2.c wire/Network.cpp sendOutbound verbatim (P1 branch live, ETH branch DEFERRED) + restore the ObBuffs ob_main call.	Audit finding: Lyra's Step-14 OutboundRing/Ep2SendThread are functionally-parallel idiom translations (vector buffers, bool+cv semaphores) predating the verbatim mandate — retired at P4 when the verbatim chain goes live.
P3	[DONE]	netInterface registrations + obbuffs ring lifecycle. SHIPPED + operator HL2 bench PASSED 2026-06-12 (RX fine; stop/start + shutdown clean) — create_rnet moved to once-per-process at the main.cpp QTimer AFTER create_xmtr (the reference C#-init order; fixes the per-open prn re-allocation leak); SendpOutboundTx(OutBound) restored at the reference site (netInterface.c:1761, tail of create_rnet); UpdateRadioProtocolSampleSize verbatim (netInterface.c:1836-1858, diff IDENTICAL) called per-open per StartAudio:45 — creates obbuffs rings 0/1 (2 new ob_main pump threads per session, idle: no producers until P4); destroy_obbuffs(0/1) at close() per StopAudio:112-113. RX side deliberately NOT registered (WdspEngine hybrid owns RX dispatch until P4 — the B.6.b clobber class).	[DONE] Operator HL2 bench PASSED 2026-06-12.
P4	[PEND]	Wire-LIVE switchover	ONE commit, full HL2 bench gate

PureSignal is a committed feature — the P0.d _cmaster struct carries the full PS surface so v0.3 lands on top without a retrofit.

Historical stage index (pre-P0 lettered stages — superseded)

The original lettered plan below is retained for history. The P0 arc absorbed/superseded it: Stage C (ilv) = P0.b; Stage D (xcmaster) = P0.d; Stage E (TxChannel audit) = obsoleted by the P0.d TxChannel deletion; Stage B's idiom-translated AAMix was re-reported verbatim at P0.c.

Stage	Status	What	Notes
A	[DONE] DONE	CMaster.cpp wire-up stubs (Send*Outbound* + SetTCIRun)	Superseded by the P0.d full verbatim CMaster port.
B	[DONE] DONE	aamix.c direct port + wire into RX path	Shipped 2026-06-08 (idiom-translated); re-reported VERBATIM at P0.c.
C	[DONE] DONE (as P0.b)	ilv.c direct port (TX I/Q interleaver)	Landed verbatim at wire/ILV.{h,cpp}.
D	[DONE] DONE (as P0.d)	xcmaster pump body direct port	Landed verbatim in wire/CMaster.cpp (real style/txid/chid + update[] CS).
E	[N/A] OBSOLETE	TxChannel reference-parity audit + reconciliation	TxChannel DELETED at P0.d — the verbatim create_xmtr replaces it.
F	[PEND] TBD	WDSP TXA chain components per CLAUDE.md §4.1 v0.2.1	compress / cfcomp / eqp / wcpagc — operator-driven per-feature direct port vs cffi-only call. Order TBD.
G	[PEND] TBD	PureSignal direct ports (v0.3)	iqc / calcc / xbuidler / delay. Lands when v0.3 PS work starts. Needs operator HL2+ with the PS hardware mod.
H	[PEND] TBD	TX/PS auto-attenuator FSM port	C# study only (PSForm.cs); Lyra-native rewrite.

Ordering note: the P0->P4 sequence above is the active linear plan. F–H land after P4 Wire-LIVE. The methodology (smallest revertible step + mechanical diff + operator bench gate) applies within each step.

OPEN QUESTIONS FOR OPERATOR

These are decisions that shape the plan but I cannot make unilaterally — surface them when picking up tomorrow:

1. **Stage C (ilv.c) vs Stage D (xcmaster) vs Stage E

(TxChannel audit) — which first?** All three are defensible:

- Stage C unlocks reference-faithful TX I/Q flow.
- Stage D removes the Lyra-native pump (cleaner end-state).
- Stage E formalizes what's already shipped (lowest risk,

highest confidence-gain).

2. **Do we re-port networkproto1.c + network.c + the Step 14

wire pieces direct from reference, OR leave them as the current mix of Lyra-native rewrites + original h12_stream.cpp** The Step 14 plan was midway through these when Stage B started. Direct-port would replace the rewrites with reference-verbatim ports. Operator call.

3. **WDSP TXA-chain components — direct port (~150-600 LOC

each) vs cffi-only? The DLL exports work; cffi is faster to ship. Direct port gives Lyra-cpp full source visibility + lets us patch behavior at the source level. Per-component operator decision; default = cffi-only unless a real source-level need surfaces.

4. **PS work timing — start v0.3 (iqc/calcc/etc.) now in

parallel with TX, or sequence after TX is fully done? The CLAUDE.md plan defers PS to v0.3 (after v0.2 TX release). Operator may want to start the ports early.

5. **The §15.27 "opt-in toggles that become defaults" cleanup

from the Python project — does it have a lyra-cpp analog here? The Python project had a long backlog of legacy toggles to retire. Lyra-cpp doesn't carry the same legacy, but new direct-port commits may obsolete some Settings surfaces operator-knows-about — track per-component.

SHIPPED-WORK LOG

Newest at top. Cross-references SESSION_LOG.md for full arc details.

2026-06-12 — P0.d cmbuffs/cmsetup VERBATIM port ([DONE] HL2 bench PASSED)

- **P0.d** afc7950 on tx-rebuild: NEW wire/cmsetup.{h,cpp}

(cmMAX* 16/4/4/2/32 + the 8 id helpers + SetRadioStructure + set_cmdefault_rates); wire/CmBuffs.{h,cpp} rewritten verbatim (cmb, *CMB, malloc0_aligned_free; Phase-C retrofit deviations removed); wire/CMaster.{h,cpp} rewritten verbatim (FULL _cmaster struct incl. the PureSignal surface, enum AudioCODEC, raw TCI fn ptrs, verbatim create/destroy_cmsetup + create/destroy_xmtr + xcmaster + all Sendp/Set* setters).

- **TxChannel RAIL carve-out DELETED** (src/wdsp/TxChannel.{h,cpp})

retired) — the verbatim create_xmtr opens the WDSP TXA channel + TX analyzer (disp 1) + out[0..2] + ILV itself through the wdspcalls seam.

- main.cpp: SetRadioStructure(2,1,1,1,0,...) config before

create_cmsetup (chid(0,0)=0 RX1 / chid(1,0)=1 TX = the live layout); create_xmtr() post-resolve_wdsp_calls (documented DEFERRED-CALLSITE accommodation); handler-1.5 = gated destroy_xmtr().

- Verification: clean build ZERO warnings; test_ilv ALL PASS;

mechanical diffs IDENTICAL (CmBuffs.{h,cpp} / cmsetup.h / CMaster.h / all 10 fully-verbatim cmsetup.c functions) + live-line-subset PASS for the partially-deferred bodies (sole accommodation: (char*)" at XCreateAnalyzer).

- New at startup: TWO cm_main pump threads (streams 0+1; stream 0

idles — no Inbound producer); TX stays wire-quiescent (OutboundTx null until P3).

- [DONE] **GATE PASSED 2026-06-12: operator HL2 RX-regression bench**

("RX still working") — P1 (obuffs.c) is unblocked.

2026-06-09 -> 2026-06-11 — P0.a/P0.b/P0.c verbatim-rewrite arc

- Operator rejected the 2026-06-09 "rule-#8" retrofit deviations

("I said PORT!") -> every ported file rewritten VERBATIM.

- **P0.a** e0c0f4a: wire/wdspcalls.{h,cpp} — the single

approved WDSP linkage seam (fn-ptr table, exact export names, PureSignal entry family complete + dumpbin-verified). scratch/_typedef_fidelity_test.cpp disproved the claimed "C++ collisions" (twin typedef / complex[2] / raw fn ptr all compile verbatim).

- **P0.b** 1f49d98: ilv.{h,c} verbatim -> wire/ILV.{h,cpp};

new wire/cmcomm.{h,cpp} (complex/PORT/malloc0/PI); pilv[] bank gone; SendpOutboundTx raw fn ptr (register AFTER create_xmtr — no null guards, reference ordering). Mechanical diff whitespace-only; test_ilv ALL PASS.

- **P0.c** e0f2584: aamix.{h,c} verbatim -> wire/AAMix.{h,cpp}

(supersedes the Stage-B idiom translation); wire/resample.h = public RESAMPLE ABI; wdspcalls retyped + flush_resample; WdspEngine outbound lambda -> static fn + TU-scope context ptr. [DONE] **Operator HL2 RX bench PASSED 2026-06-11.**

2026-06-08 EOD — Stage B aamix.c port COMPLETE

- **Stage B (aamix.c)** [DONE] direct port shipped + bench-validated
- Commits 27ac2fa -> 7a50b07 on tx-rebuild
- Operator HL2+ bench 19:27:49 -> 19:28:47: audible RX through

both AK4951 jack and PC-Soundcard sinks; mute/unmute via dispatch; output-device swap; AGC mode changes; chain out=N tracks 1:1 with dispatch calls=N across both sinks

- B.6.b root cause: Stage-A SendpOutboundRx no-op stub at

RadioNet.cpp:272 clobbered aaMix_->Outbound ~1 ms after WdspEngine::openRx1 wired the real lambda. Fixed by deleting the stub (commit 8b8e0da).

- 2-stage diagnostic instrumentation (chain @ 225518c +

dispatch @ 12369bc) localized the defect to a single line of code on the 2nd bench — pure-speculation cost avoided.

- Methodology lessons locked into this doc § "LOCKED

METHODOLOGY" above (counters-first when symptom is "nothing happens"; operator-empirical rule held; "do as Thetis does" continues to pay).

- Tasks #130 (parent) + #132 (Stage B.6.b) closed.
- Backup _backups/lyra-cpp-2026-06-08-aamix-stage-B-COMPLETE-with-docs.bundle.

Earlier work

See SESSION_LOG.md for the full session-by-session log. The Step 14 wire-layer arc (Stages 1, 1.5, 2a, 2b1, 2b2) is covered in detail there + in EXECUTION_PLAN.md's progress- tracking sections. TX-1 components 2-6 (main branch) are covered in project_lyra_cpp_tx.md (memory file).

RESUME POINTER (next session)

First actions (locked discipline):

1. git fetch origin
2. git status + git log --oneline -5 tx-rebuild origin/main
3. Read this doc + the "READ FIRST" block in project_lyra_cpp_tx.md
4. Read SESSION_LOG.md newest entry for arc detail

Then: wait for operator decision on which Stage (C / D / E / F / open-questions item) to land next. Each Stage gets its own grounded design + reference-read + smallest-revertable- step + bench-gated implementation pattern matching the Stage B arc.

Last updated: 2026-06-08 EOD — Stage B aamix.c port shipped + bench-validated. Next port target = operator's call.